Performance Improvement of Electrostatic Actuator by Skewing Electrodes

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Abstract **The authors have developed several types of high power electrostatic actuators. DEMED (Dual Excitation Multiphase Electrostatic Drive) is one of those actuators and features a higher power/weight ratio than conventional electromagnetic motors. DEMED is expected to replace the conventional motors in applications that require high-power or light weight. For those applications, DEMED has sufficient performance in terms of power generation, however, its output is very rippled. This paper proposes the use of the skewing technique, which is commonly used in electromagnetic motors, to DEMED electrodes in order to reduce the ripple of force. Some analytical results along with fundamental experimental results are presented. The analysis indicates that appropriately skewed electrodes can reduce the ripple of force considerably and also can increase the practical force of DEMED.**

I. INTRODUCTION

Nowadays, the need for new types of drives which have higher performance than conventional motors is increasing, as mechanical systems are progressing. To satisfy those demands, the authors have developed several types of high-power and light-weight electrostatic actuators [1..3]. Among them, DEMED (Dual Excitation Mutiphase Electrostatic Drive) has the best performance in terms of power generation. The DEMED technology was applied to a linear motor and a prototype, weighing 7.0g, generated a power of 1.6W. Its power/weight ratio, 230W/kg, is much larger than typical electromagnetic motors. DEMED technology can also be applied to rotary motors and a prototype has been built [3].

DEMED can be used in many applications that require high-power generation or light-weight. Although DEMED has high performance in power, it is very hard to control DEMED precisely because of the strong ripple in the output torque or thrust force. DEMED has also some problems related with vibration and noise. The main cause of these problems are considered to lie in the ripple of the output, hence, suppression of the ripple will result in a removal of the mentioned problems.

In electromagnetic motors, the torque ripple can be reduced by an appropriate design of the structure. The most widespread method is by skewing either the stator slots or rotor magnets. Considering the analogy between magnetic motors and electrostatic motors, skewing the electrodes of DEMED is estimated to reduce the ripple of force. This paper describes the effect of skewing the electrodes of a linear type DEMED.

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II. DEMED

In this section, a linear type of DEMED is described. Static characteristics of DEMED are presented and some parameters which represent the characteristics are defined.

A. Operation Principle

A fundamental DEMED unit consists of a pair of plastic films – slider and stator – as shown in Fig. 1. Each film contains 3-phase parallel electrodes. By applying 3-phase ac voltages on these 3-phase electrodes, traveling potential distributions are generated on the two films, as shown in Fig. 2. The interaction between the two field distributions yields a propulsive electrostatic force, which drives the slider in the direction shown in Fig. 1 and 2. The field interaction also causes a moment which suppresses the yawing motion of the slider. Therefore, usually no guiding element, such as a linear guide, is required. The synchronized slider velocity, u, is expressed as follows:

$$u = 6\,p\,v \qquad (1)$$

where p is the pitch of the electrodes and v is the frequency of the applied ac voltages.

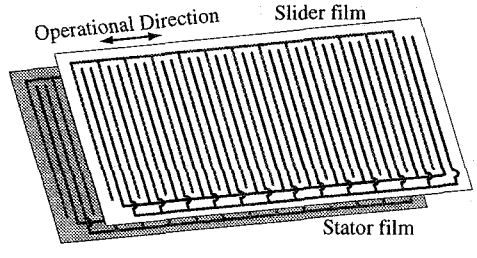


Fig. 1 Basic unit of DEMED

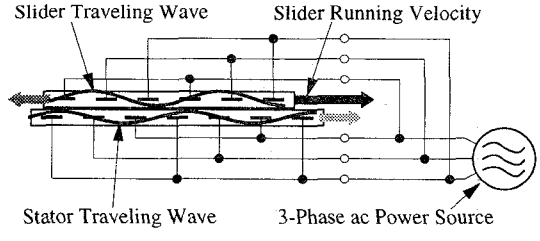


Fig. 2 Cross-sectional schematic view of the operation principle of DEMED

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B. Static Characteristics

Fig. 3 shows the static force characteristics of DEMED which the authors have formerly obtained analytically [3]. In this figure, the characteristics are expressed as a function of the phase of the applied ac voltage and of the slider position. To evaluate the characteristics, the *synchronization path*, *pull-out force*, and *force ripple* are defined.

Fig. 4 is the contour plot of Fig. 3. The light areas represent positive propulsion forces while the dark areas represent negative propulsion forces. For ease of analysis, the motor's operation under a certain load is considered to proceed approximately along a straight line and is termed as *synchronization path*. Each synchronization path is characterized by a representative force which is determined by its minimum value along the path.The greatest representative force is defined as *pull-out force* which expresses the maximum attainable thrust force of the motor.

The *force ripple* is defined as:

$$\text{force ripple} = f_d / f_m \tag{2}$$

where f_d is the difference between the maximum and minimum force along the synchronization path that represents the pull-out force and f_m is the mean force along the path.

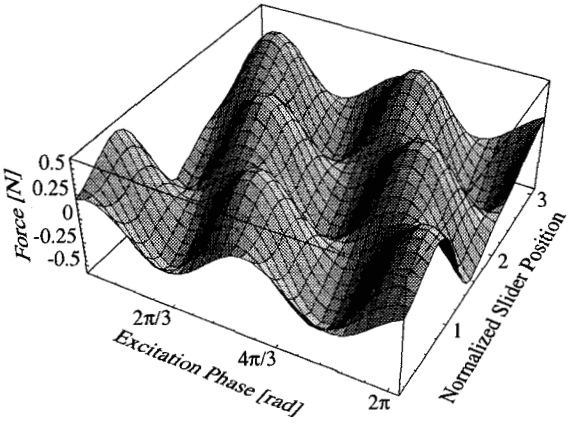


Fig. 3 Static propulsive force characteristics of DEMED calculated by Surface Charge Method

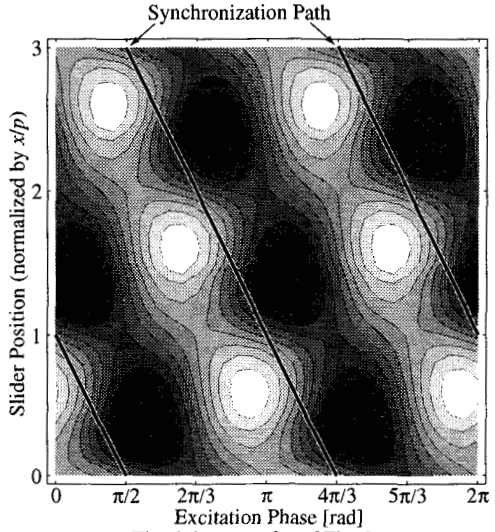


Fig. 4 Contour plot of Fig. 3

III. Skewing Electrodes

Fig. 5 shows the schematic view of the skewed electrodes. The slider electrodes are rotated while maintaining an uniform pitch p to form an angle θ with the stator electrodes, as shown in Fig. 5. Since the slider and stator are symmetrical, the slider's line of driving direction bisects the two lines which are perpendicular to the stator and to the slider electrodes, respectively. The resulting intervals of both the stator and slider electrodes in the driving direction are termed as *apparent pitch* p_a. The relationship between p and p_a is expressed as:

$$p = p_a \cos\left(\frac{\theta}{2}\right) \tag{3}$$

DEMED moves periodically with steps of p_a.

To express the degree of skewing, the *skew factor* F_s is introduced and is defined as:

$$F_s = \frac{l \sin\theta}{p} \tag{4}$$

where l is the length of each slider electrode. The skew factor defines the number of stator electrodes crossed by each slider electrode.

In the following discussion, the DEMED based on skewed electrodes is called DEMED-s.

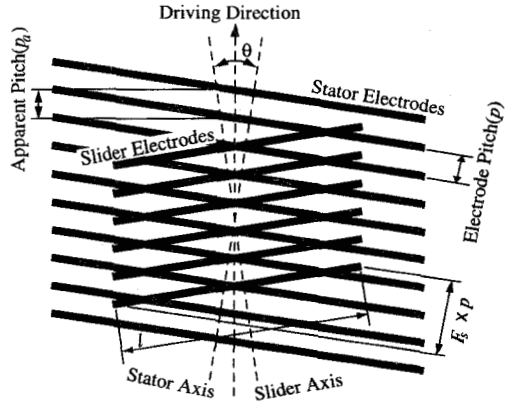


Fig. 5 Schematic top view of skewed electrodes of DEMED-s

IV. Analysis

To evaluate the effect of skewing, the static force characteristics of DEMED-s is analyzed. As mentioned above, the conventional, without skewed electrodes, DEMED can operate without any guiding elements, such as linear guides, due to the suppressive action of the moment on the yawing motion of the slider. However, depending on the skew factor the moment exerted on the slider can destabilize the slider's motion. In the following discussion, the moment which suppresses the yawing motion of the slider is called restriction moment.

A. Theoretical Analysis

The characteristics of DEMED-s is derived from the static force characteristics of the conventional DEMED, shown in Fig. 3 and 4. The thrust force of the DEMED-s, f_{Fs}, is given by the following equation:

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$$f_{F_s}(\phi,x_a) = \int_{x_a-\frac{F_s}{2}}^{x_a+\frac{F_s}{2}} \frac{f(\phi,x)}{F_s}\,dx \qquad (5)$$

where $f(\phi, x)$ denotes the generated force of the conventional DEMED, ϕ is the phase of the applied ac voltage, x is the slider position normalized by p in case of the conventional DEMED, and x_a denotes the position of the slider's geometrical center in the DEMED-s with respect to the stator and is normalized by p_a.

The moment acting on the slider, m_{Fs}, can be derived from the following equation:

$$m_{F_s}(\phi,x_a) = \int_{x_a-\frac{F_s}{2}}^{x_a+\frac{F_s}{2}} \frac{(x-x_a)\,f(\phi,x)}{F_s}\,dx \qquad (6)$$

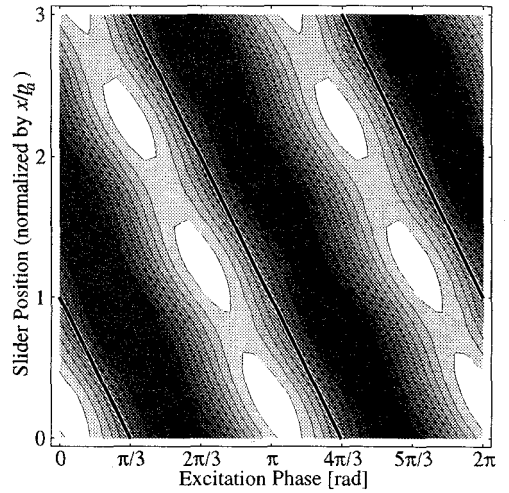
B. Results and Discussion

As an example, Fig. 6 depicts the force and moment characteristics for $F_s = 1.5$. In Fig. 6(b), the moment increases the angle θ of the slider in the light (positive) areas and decreases it in the dark (negative) areas. Fig. 7(a) and 7(b) show the pull-out force and force ripple as a function of F_s which can be derived from the force characteristics of each skew factor. The maximum force in Fig. 7(a) expresses the maximum value of f_{FS}.

To evaluate the effect of skewing on the moment, the following strategy is used. The bold lines drawn in Fig. 6(b) represent the synchronization paths for the case of no load which can be extracted from the force characteristics (Fig. 6(a)). The moment characteristics is represented by the representative moment which is defined as the mean moment over this path. The relationship between the representative moment and skew factor is shown in Fig. 7(c).

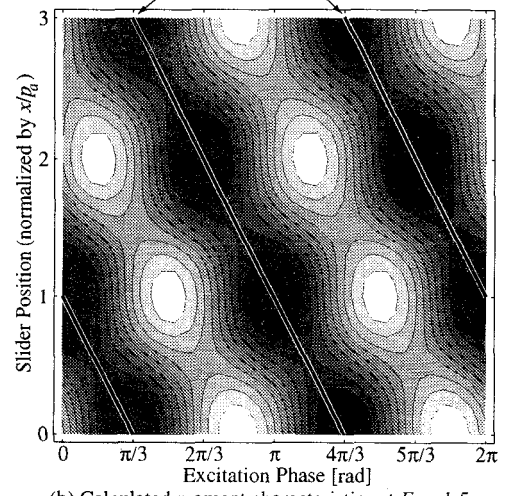
As shown in Fig. 7(a), the maximum force is decreasing in the skew factor region between zero and three. On the other hand, the pull-out force has a global maximum near $F_s = 1.5$. The reason for this is the fact that the minimum value of the force along the synchronization path that represents the pull-out force is raised by averaging forces. For values of F_s greater than 1.5, the pull-out force shows up local maxima periodically for $F_s \cong 3k+1.5$, $k>0$, where k is an integer.

The force ripple possesses local minima at $F_s \cong 3k\text{-}1.5$, $k>0$, where k is an integer, which are located near to the values of F_s for which the pull-out force has its maxima. For these skew factors the force ripple has a value of approximately a quarter of that at $F_s = 0$, therefore, effective ripple reductions can be expected. With respect to the moment, zero values are attained at the same skew factors for which the force ripple has its minima



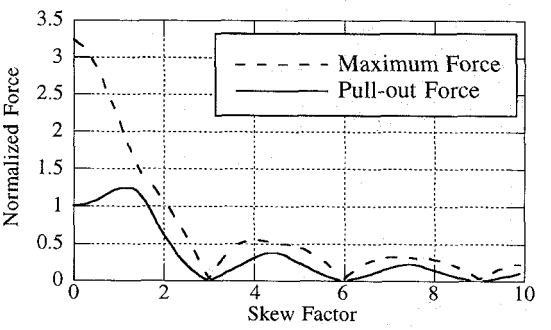
(a) Calcualted force charactersitics at $F_s = 1.5$

Synchronization Path
derived from the force characteristics of the same skew factor

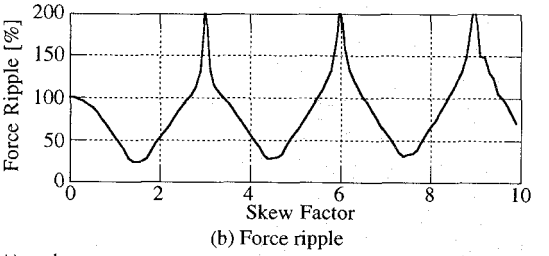


(b) Calculated moment characteristics at $F_s = 1.5$

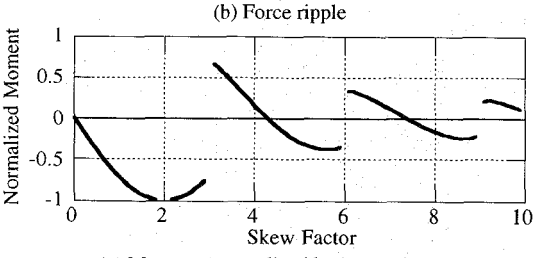
Fig. 6 Calculated force and moment characteristics at $F_s = 1.5$



(a) Force (normalized by pull-out force at $F_s = 0$)



(b) Force ripple



(c) Moment (normalized by its maximum)

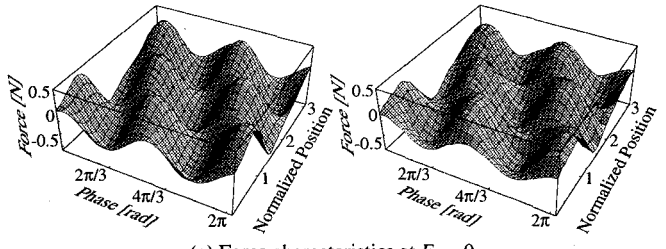
Fig. 7 Effects of skewing the electrodes of DEMED

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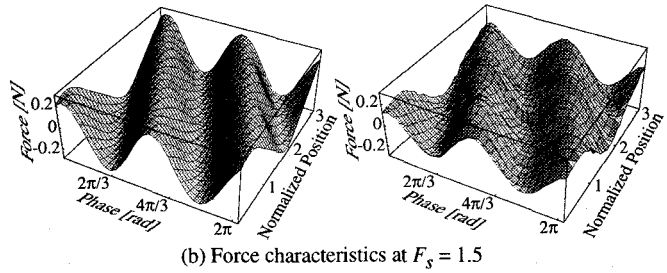
with the exception of F_s = 1.5. Near these skew factors where the moment becomes zero, the moment acts as a restriction moment.

C. Measurements

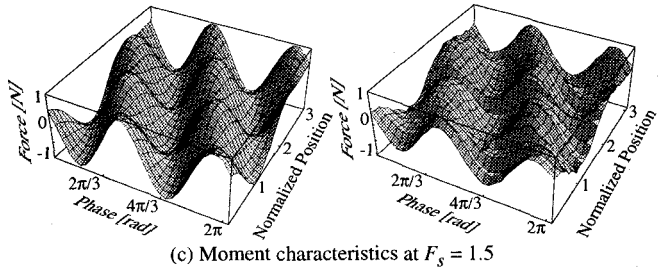
To confirm the accuracy of the analysis, measurements of the static characteristics are performed for different values of F_s and compared with the calculated ones. In the experimental setup, the slider and the stator consist of glass-epoxy plates on which the electrodes are produced using conventional etching techniques. A 6-axes force sensor attached to the slider measures the force and moment acting on the slider. The experimental results are shown in Fig. 8 along with the analytical results. As evidenced from Fig. 8, the analytical results agree well with the experimental results obtained for F_s = 0 and 1.5. By calculating the pull-out force from the experimentally obtained force distributions, it is confirmed that the pull-out force at F_s = 1.5 is almost 20% higher than the case where no skewing was used, as shown in Fig. 7.



(a) Force characteristics at F_s = 0



(b) Force characteristics at F_s = 1.5



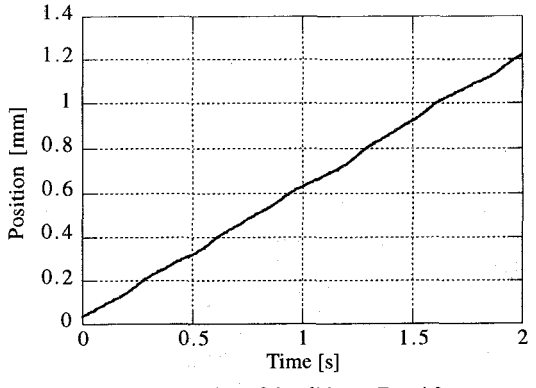
(c) Moment characteristics at F_s = 1.5

Fig. 8 Calculated (left) and measured (right) characteristics

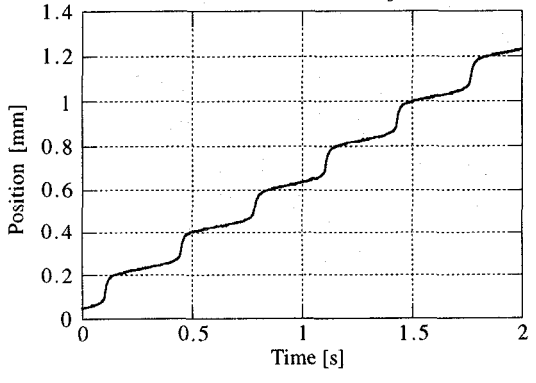
V. Ripple-less Drive of DEMED

At the skew factors given by $F_s \cong 3k+1.5$, $k>0$, where k is an integer, DEMED can operate easily and smoothly without the use of linear guides. The slider motion of a prototype DEMED having a F_s = 4.3 was measured. For this purpose, an electro-optical displacement sensor (ZIMMER MODEL 200) was utilized. By attaching a target on the geometrical center of the slider, the slider motions at F_s = 0 and F_s = 4.3 were measured as shown in Fig. 9. According to Fig. 7(a), the generated force at F_s = 0 is almost three times as large as that at F_s = 4.3. The amplitude of the slider motion ripple varies with the generated force. Therefore, the applied voltage at F_s = 0 is lowered compared to that at F_s = 4.3 in order to compare the amplitude of the ripples on equal terms. As can be observed, the slider motion at F_s = 4.3 is very smooth in contrast to the motion where no skewing was used, which confirms an effective reduction in the ripple.

Although the motion at F_s = 4.3 is very smooth, as shown in Fig. 9, the slider possesses a yawing motion to some extent. Fig. 10 shows the motion of the corner of the slider at F_s = 4.3. The motion of the corner has a ripple larger than that of the geometrical center, resulting in the yawing motion of the slider



(a) Motion of the slider at F_s = 4.3



(b) Motion of the slider at F_s = 0

Fig. 9 The motion of the slider at F_s = 4.3 and F_s = 0
The electrode pitch is 200μm, frequency of the ac voltage is 0.5Hz.
The excitation voltage is 2200V_{rms} at F_s = 4.3, and 1225V_{rms} at F_s = 0.

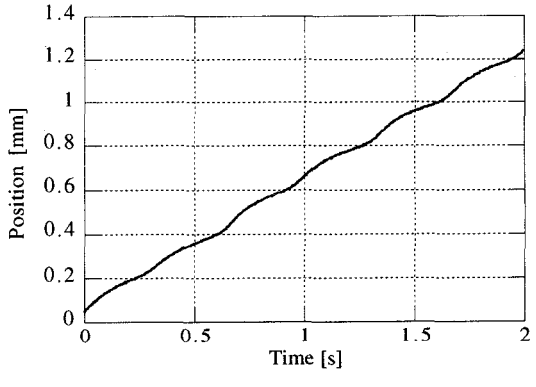


Fig. 10 Motion of the corner of the slider at F_s = 4.3
The excitation voltage is 2200V_{rms}

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as illustrated in Fig. 11.

Fig. 12 shows the calculated moment characteristics at F_s = 4.3. The bold lines represent the synchronization paths for no load which are obtained from the force characteristics at F_s = 4.3. From this figure, it is found that the moment at F_s = 4.3 alternates between positive and negative values periodically. This characteristic agrees with the measured motion shown in Fig. 11.

In this section, only the characteristics at F_s = 4.3 was discussed. However, the same observations can be made for skew factors given by $F_s \cong 3k+4.5$, $k>0$, where k is an integer.

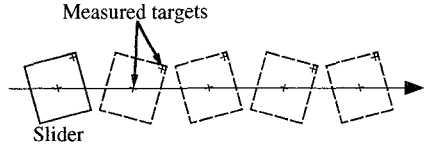


Fig. 11 Estimated motion of the slider at F_s = 4.3

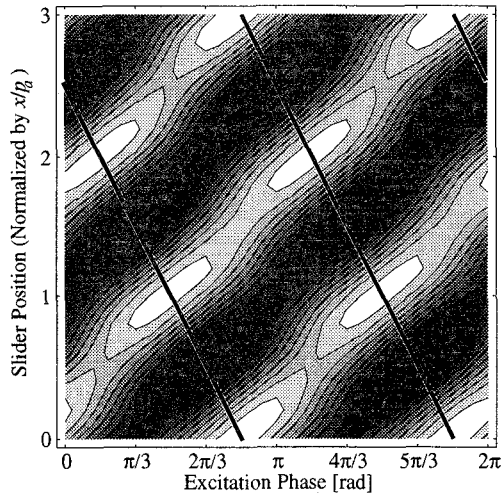


Fig. 12 Moment characteristics at F_s = 4.3
Bold lines represent synchronization paths derived from force characteristics of the same skew factor

VI. V-DEMED

As mentioned above, by utilizing skew factors $F_s \cong 3k+1.5$, $k>0$, where k is an integer, the force ripple is minimized resulting in a smooth slider motion. However, for these skew factors the generated force is smaller than that without the use of skewing, if the same voltages are applied. On the other hand, at F_s = 1.5, the pull-out force possesses a global maximum and the ripple is minimized (Fig. 7). However, the yawing motion is unstable at F_s = 1.5 , due to the moment acting on the slider. Therefore, a skew factor of 1.5 can be considered as most effective provided that a remedy for the destabilizing moment acting on the slider is found.

The use of a V-shaped electrode pattern is proposed as a solution to this problem, which is shown in Fig. 13. The slider electrodes are arranged to form a V-shaped pattern having a skew factor F_s = 1.5 whereas the stator electrodes remain straight. We call this type of DEMED, V-DEMED. As an alternative, it is possible to use a combination of V-shaped stator electrodes and straight slider electrodes. In the following

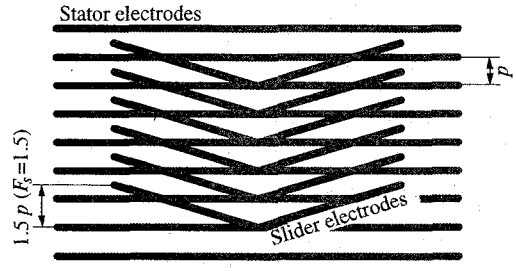


Fig. 13 Schematic top view of electrodes of V-DEMED

discussion it is assumed that the slider has a V-shaped electrode pattern and the stator has normal straight electrodes. The force and moment of V-DEMED is derived from the following equation:

$$f_V(\phi,x_v) = \int_{x_v}^{x_v+F_{s_1}} \frac{f(\phi,x)}{F_{s_1}}\,dx + \int_{x_v}^{x_v+F_{s_2}} \frac{f(\phi,x)}{F_{s_2}}\,dx \qquad (7)$$

$$m_V(\phi,x_v) = \int_{x_v}^{x_v+F_{s_1}} \frac{(x-x_v)\,f(\phi,x)}{F_{s_1}}\,dx - \int_{x_v}^{x_v+F_{s_2}} \frac{(x-x_v)\,f(\phi,x)}{F_{s_2}}\,dx \qquad (8)$$

where x_v is the position normalized by p of the geometrical center of the slider with respect to the stator, and the other variables are defined as in Fig. 14. When the slider is positioned at an angle θ =0, as shown in Fig. 13, no moment acts on the slider. It is important to know how the sign of the moment varies when the slider yaws with an angle $\theta \neq 0$. Fig. 15 shows a graph of the moment exerted on the slider for a yawing angle of 0.05°. The bold lines represent the synchronization paths for zero load and are derived from the force distribution. The force distribution is almost the same as that in Fig. 6(a), however, a shift of the vertical axis can be observed as the shape of the slider electrode pattern changes. In Fig. 15, restriction moments act on the slider in the dark (negative) areas, and destabilizing yawing moments act in the light (positive) areas. This implies that V-DEMED possesses a positive stiffness for yawing motion in the dark areas and a negative stiffness in the light areas. The rotational stiffness, k_r, is defined as:

$$k_r = -\frac{\partial m_r}{\partial \theta} \qquad (9)$$

where m_r is the mean moment over the synchronization paths and θ is the yawing angle of the slider.

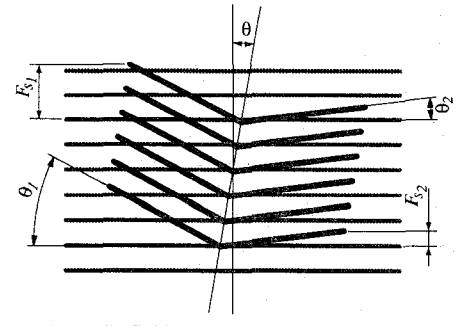


Fig. 14 Definition of variables in V-DEMED

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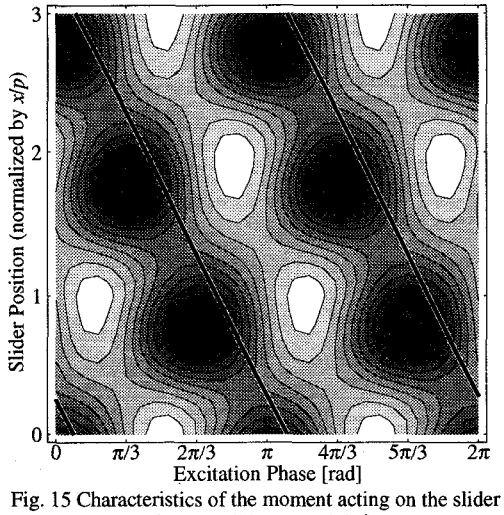


Fig. 15 Characteristics of the moment acting on the slider when the slider yaws 0.05°

For each synchronization path, the representative force and the rotational stiffness can be calculated. Fig. 16 shows the rotational stiffness and the representative force for every synchronization path that lies in the region bounded by the two synchronization paths which represent the positive and negative pull-out force, respectively. In Fig. 16, the path for which the average force becomes zero is defined as the reference path. The path number is defined as the distance to the reference path in vertical direction. As noted above, the V-DEMED cannot operate stably if the stiffness along the synchronization path is negative. The gray area in Fig. 16 represents the unstable region where the stiffness is negative. It is found from Fig. 16 that V-DEMED can operate stably when it generates positive forces (positive direction is upward in Fig. 13), whereas an unstable operation exists when it generates negative forces exceeding 60% of the maximum representative force.

Fig. 17 shows the representative force and stiffness of a conventional DEMED (no skewing, parallel straight electrodes) calculated under the same conditions (yaw of the slider, electrode length, electrode pitch) as in Fig. 16. As can be seen, the conventional DEMED cannot generate the maximum representative force, that is pull-out force, when the yawing motion is required to be stable. Therefore, it is concluded that V-DEMED can generate a 35% larger force than a conventional DEMED if the rotational stiffness is restricted to be positive. Naturally, it is possible to operate DEMED in the unstable area by using linear guides, however, in that case the inherent compactness of DEMED will be undone. In V-DEMED the motion characteristics varies with the direction of the force generation as shown in Fig. 16. However, this variation can be reduced by using multi-V shaped electrodes, e.g., W-shaped electrodes.

VII. Conclusion

In this paper, the effect of skewing electrodes in DEMED was theoretically and experimentally analyzed. Both results of analyses and experiments agreed very well. The analytical results show that for appropriate skewing factors given by $F_s \cong 3k+1.5$, $k>0$, where k is an integer, a considerable reduction in the force ripple of DEMED can be accomplished. That was confirmed by the experimentally measured motions.

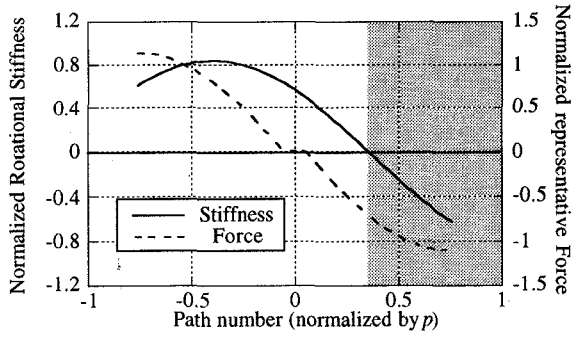


Fig. 16 Stiffness and pull-out force of V-DEMED
Pull-out force and stiffness are normalized by their maximum values in a conventional DEMED as shown in Fig. 17

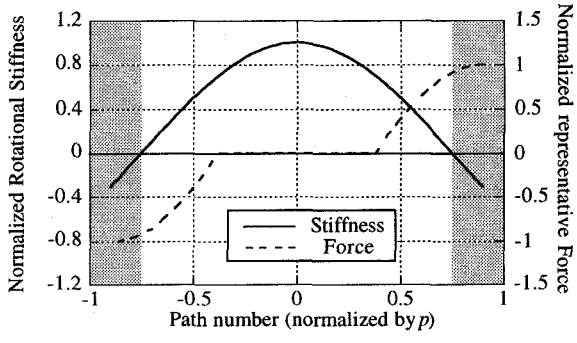


Fig. 17 Stiffness and the pull-out force of normal DEMED
Pull-out force and stiffness are normalized by their maximum values

It was also shown that an increase in the pull-out force and decrease in ripple can be obtained by using a V-shaped electrode pattern having a skewing factor of $F_s = 1.5$. As future research, it is necessary to verify these results experimentally with a prototype V-DEMED.

The skewing method presented in this paper will make DEMED overcome its defects such as noise, vibration, and difficulties of precise motion control. By using the methods proposed in this paper, particularly V-DEMED, many applications of DEMED will be realized.

Reference

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